

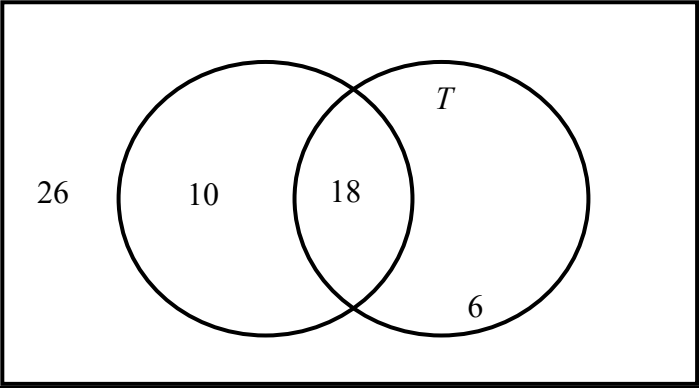
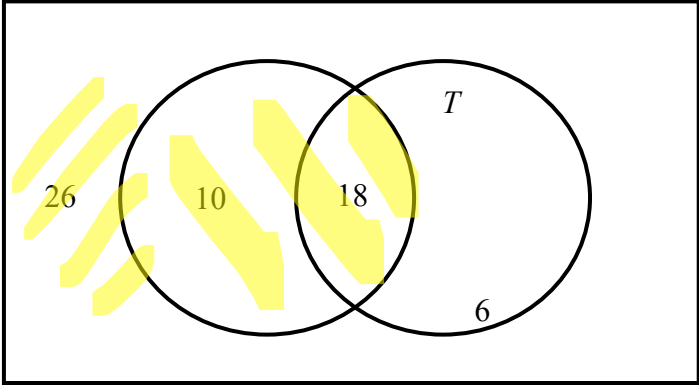
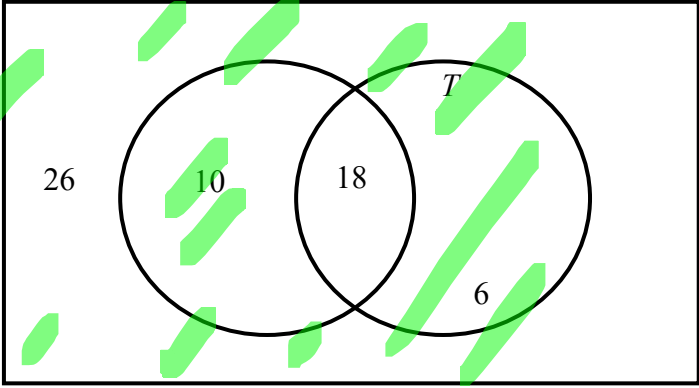
**Bedok South Secondary School**  
**Mathematics Department**  
**Secondary 4 Exp Prelim P1 2026**  
**Solutions**

| Question |   | Solution                                                                                                                                                                                                            |
|----------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        |   | $\frac{\sqrt{899} \times 0.197}{\pi}$ $\approx \frac{\sqrt{900} \times 0.2}{3}$ $= 2$                                                                                                                               |
| 2        | a | $3528 = 2^3 \times 3^2 \times 7^2$                                                                                                                                                                                  |
|          | b | $3528 \times k = 2^3 \times 3^2 \times 7^2 \times k$<br>To be perfect cube, all powers must be 0 or <b>multiples of 3</b> .<br>$k = 3 \times 7 = 21$                                                                |
| 3        |   | 81% of original price = 647<br>original price = $647 \div 0.81$<br>$= 798.7654$<br>$\approx 799$                                                                                                                    |
| 4        |   | Let the initial number of boys and girls be $16u$ and $11u$ respectively<br>At the end,<br>$\frac{16u + 23}{11u - 15} = \frac{7}{3}$ $48u + 69 = 77u - 105$ $29u = 174$ $u = 6$ Number of girls = $11(6) - 15 = 51$ |

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| <b>5</b> | <b>a</b> | $-66 \leq 12(7-9x) < 57$<br>$-66 \leq 12(7-9x)$ and $12(7-9x) < 57$<br>$-66 \leq 84-108x$ and $84-108x < 57$<br>$108x \leq 150$ and $27 < 108x$<br>$x \leq \frac{25}{18}$ and $\frac{1}{4} < x$<br>$\frac{1}{4} < x \leq \frac{25}{18}$                                                                                                                                                                                                                                                                                                                               |
|          | <b>b</b> | $(3a-2b)(2a+5b)$<br>$= 6a^2 + 15ab - 4ab - 10b^2$<br>$= 6a^2 + 11ab - 10b^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>6</b> |          | $\frac{x^2-9}{2x^2+5x-3} \div \frac{x-9}{2x-1}$ $= \frac{x^2-9}{2x^2+5x-3} \times \frac{2x-1}{x-9}$ $= \frac{(x-3)(x+3)}{(2x-1)(x+3)} \times \frac{2x-1}{x-9}$ $= \frac{x-3}{x-9}$                                                                                                                                                                                                                                                                                                                                                                                    |
|          |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>7</b> |          | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math>\angle ACB = 180^\circ - 148^\circ = 32^\circ</math><br/> <math>\sin 32^\circ = \frac{AB}{83}</math><br/> <math>AB = 83 \sin 32^\circ</math><br/> <math>= 43.98329</math><br/> <math>\approx 44.0 \text{ m}</math> </div> <div style="width: 45%;"> <b>OR</b><br/> <math>\sin \angle ACB = \sin(180^\circ - \angle ACB)</math><br/> <math>\frac{AB}{83} = \sin 148^\circ</math><br/> <math>AB = 43.98329^\circ</math><br/> <math>\approx 44.0^\circ</math> </div> </div> |

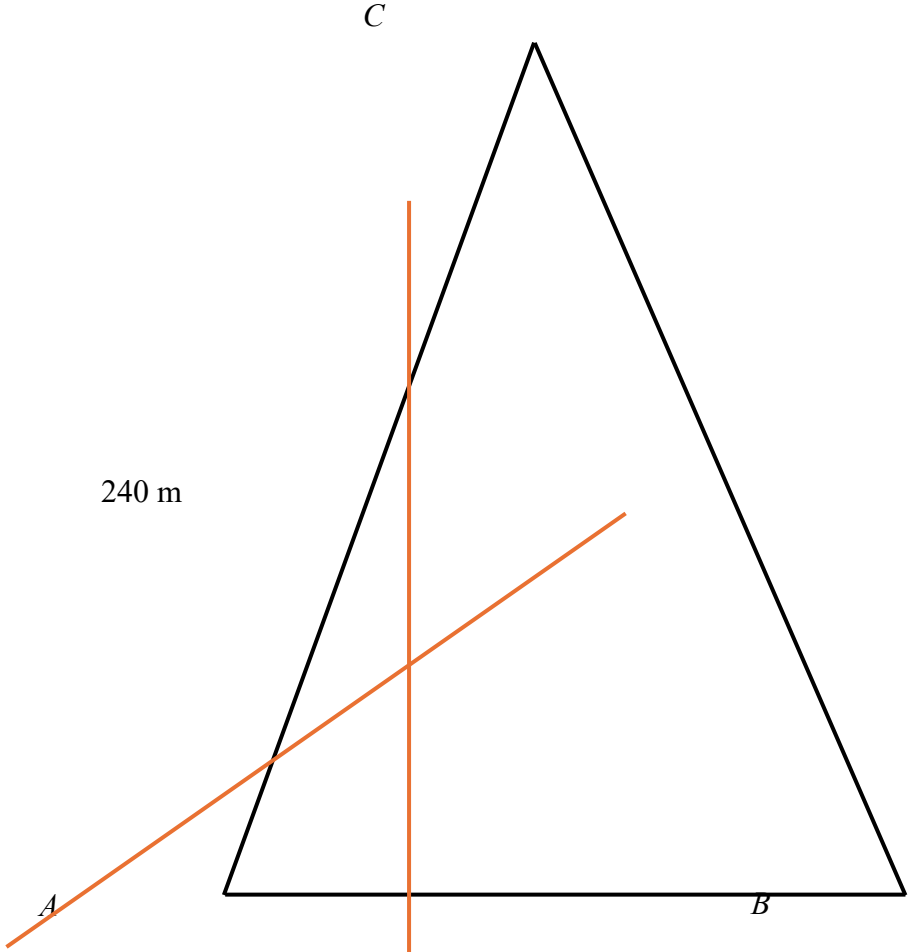
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| <b>8</b> | <b>a</b> | <div> <math display="block">A = (5x+10)(2x+4) - (x+2)^2</math> <math display="block">A = 5(x+2) \times 2(x+2) - (x+2)^2</math> <math display="block">A = 10(x+2)^2 - (x+2)^2</math> <math display="block">A = 9(x+2)^2</math> </div> <div> <b>OR</b> <math display="block">A = (5x+10)(2x+4) - (x+2)^2</math> <math display="block">A = 10x^2 + 20x + 20x + 40 - x^2 - 4x - 4</math> <math display="block">A = 9x^2 - 36x + 36</math> <math display="block">A = 9(x^2 - 4x + 4)</math> <math display="block">A = 9(x+2)^2</math> </div>                                                                                                                                                                                                                                                                                         |
|          | <b>b</b> | $A = 9(x+2)^2$ $\frac{A}{9} = (x+2)^2$ $x+2 = \pm \sqrt{\frac{A}{9}}$ $x = -2 + \sqrt{\frac{A}{9}} \text{ (rej. -ve)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|          | <b>c</b> | <div> <p>Since all dimensions are integers, all lengths are at least 1</p> <math display="block">x+2 \geq 1</math> <math display="block">(x+2)^2 \geq 1</math> <math display="block">9(x+2)^2 \geq 9</math> <math display="block">A \geq 9</math> </div> <div> <b>OR</b> <math display="block">x \geq -1</math> <math display="block">-2 + \sqrt{\frac{A}{9}} \geq -1</math> <math display="block">\sqrt{\frac{A}{9}} \geq 1</math> <math display="block">\frac{A}{9} \geq 1</math> <math display="block">A \geq 9</math> </div> <div> <b>OR</b> <p>(minimum) <math>x+2 = 1</math></p> <math display="block">x = -1</math> <math display="block">-2 + \sqrt{\frac{A}{9}} = -1</math> <math display="block">\sqrt{\frac{A}{9}} = 1</math> <math display="block">\frac{A}{9} = 1</math> <math display="block">A = 9</math> </div> |

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|    |     |                                                                                                                                        |                                                                                                                         |           |                      |
| 9  | ai  |                                                                                                                                        | Monthly Electricity Consumption, $x$ (kWh)                                                                              | Mid-Value | Number of households |
|    |     |                                                                                                                                        | $150 \leq x < 250$                                                                                                      | 200       | 9                    |
|    |     |                                                                                                                                        | $250 \leq x < 400$                                                                                                      | 325       | 26                   |
|    |     |                                                                                                                                        | $400 \leq x < 500$                                                                                                      | 450       | 20                   |
|    |     |                                                                                                                                        | $500 \leq x < 700$                                                                                                      | 600       | 18                   |
|    |     |                                                                                                                                        | $700 \leq x < 1000$                                                                                                     | 850       | 7                    |
|    |     |                                                                                                                                        | $\text{Mean} = \frac{9(200) + 26(325) + 20(450) + 18(600) + 7(850)}{9 + 26 + 20 + 18 + 7}$ $= \frac{36000}{80}$ $= 450$ |           |                      |
|    | aii | SD = 176.555 $\approx$ 177                                                                                                             |                                                                                                                         |           |                      |
|    | b   | The data is continuous and not discrete. <b>OR</b><br>The class intervals are unequal.                                                 |                                                                                                                         |           |                      |
|    |     |                                                                                                                                        |                                                                                                                         |           |                      |
| 10 | a   | $2x + 8 = 2(y + 2)$ ----- (1)<br>$2(x + 4) = 2(y + 2)$<br>$x + 4 = y + 2$<br>$x = y - 2$                                               |                                                                                                                         |           |                      |
|    | b   | $2x + 8 = 10 + y + 2$ ----- (2)<br>Sub $x = y - 2$ into (2):<br>$2(y - 2) = 12 + y$<br>$2y - 4 = 12 + y$<br>$y = 8$<br>$x = 8 - 2 = 6$ |                                                                                                                         |           |                      |

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| 11 | a | $\xi$                                                                                                                                                                                    |
|    | b | $\xi$  <p> <math>n[(I \cap T') \cup T] = 26 + 10 + 18 = 54</math> </p>                                                                                                                   |
|    | c | $\xi$  <p>             The set represents the set of students with <b>both</b> an Instagram and TikTok account <b>and also</b> the set of students with <b>neither</b>.           </p> |

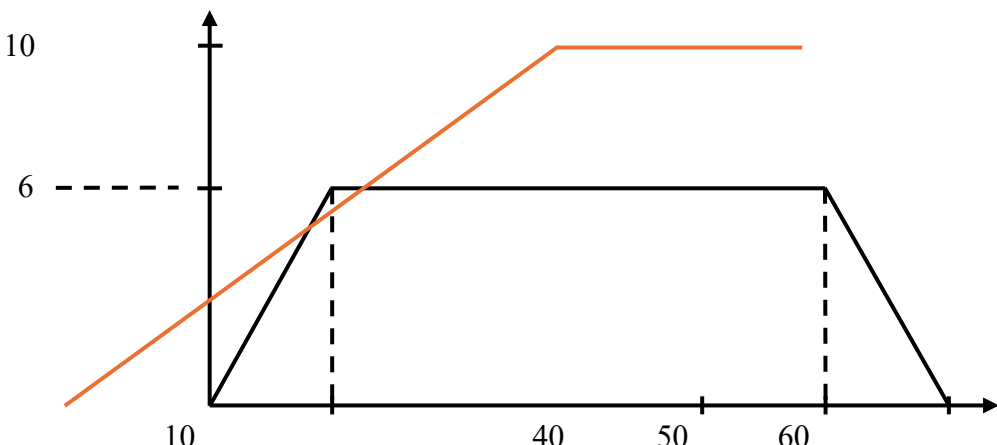
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|           |          |                                                                                                                                                                                                                                                                                                                            |
| <b>12</b> | <b>a</b> | $\frac{x}{x-2} - \frac{2}{x+1} - \frac{x+4}{(x-2)(x+1)}$ $= \frac{x(x+1)}{(x-2)(x+1)} - \frac{2(x-2)}{(x+1)(x-2)} - \frac{x+4}{(x-2)(x+1)}$ $= \frac{x(x+1) - 2(x-2) - (x+4)}{(x-2)(x+1)}$ $= \frac{x^2 + x - 2x + 4 - x - 4}{(x-2)(x+1)}$ $= \frac{x^2 - 2x}{(x-2)(x+1)}$ $= \frac{x(x-2)}{(x-2)(x+1)}$ $= \frac{x}{x+1}$ |
|           | <b>b</b> | $\frac{2x-1}{3} - \frac{x+4}{2} = \frac{5}{6}$ $2(2x-1) - 3(x+4) = 5$ $4x - 2 - 3x - 12 = 5$ $x = 19$                                                                                                                                                                                                                      |
| <b>13</b> |          | $17107.33 = 15000 \left( 1 + \frac{r \div 2}{100} \right)^{3 \times 2}$ $\frac{17107.33}{15000} = \left( 1 + \frac{r}{200} \right)^6$ $\sqrt[6]{\frac{17107.33}{15000}} = 1 + \frac{r}{200}$ $r = 200 \left( \sqrt[6]{\frac{17107.33}{15000}} - 1 \right)$ $r = 4.43024...$ $\approx 4.43$                                 |
| <b>14</b> |          | $\frac{16a^5b^{-3}}{D} = 2a^2b$ $D = \frac{16a^5b^{-3}}{2a^2b}$ $= 8a^3b^{-4} = \frac{8a^3}{b^4}$                                                                                                                                                                                                                          |

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| 15 |  | $\begin{aligned}\overrightarrow{BC} &= \overrightarrow{BA} + \overrightarrow{AC} &  \overrightarrow{BC}  &= \left  \begin{pmatrix} -6 \\ c-2 \end{pmatrix} \right  \\ &= -\overrightarrow{AB} + \overrightarrow{AC} & \sqrt{45} &= \sqrt{(-6)^2 + (c-2)^2} \\ &= -\begin{pmatrix} 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ c \end{pmatrix} & 45 &= 36 + c^2 - 4c + 4 \\ &= \begin{pmatrix} -6 \\ c-2 \end{pmatrix} & c^2 - 4c - 5 &= 0 \\ & & (c-5)(c+1) &= 0 \\ & & c &= 5 \text{ or } c = -1\end{aligned}$ <p style="text-align: right;"><b>OR</b></p> $\begin{aligned}45 &= 36 + (c-2)^2 \\ (c-2)^2 &= 9 \\ c-2 &= \pm 3 \\ c &= 5 \text{ or } -1\end{aligned}$ |
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| 16 | a<br>b |                                                                                                                                                   |
|    | c      | <p>Length from <math>A</math> to intersection = 5.5 cm</p> <p>Distance from <math>A</math> to intersection <math>5.5 \times 20 = 110</math></p> <p>Angle of elevation</p> $= \tan^{-1} \left( \frac{50}{110} \right)$ $= 24.4^\circ$ |

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| 17 | a | $\text{Int } \angle \text{ of octagon} = \frac{(8-2) \times 180}{8}$ $= 135$ $\text{Int } \angle \text{ of } n\text{-gon} = 360 - 81 - 135 = 144$ $(n-2) \times 180 = 144n$ $180n - 360 = 144n$ $36n = 360$ $n = 10$ <div style="text-align: right;"> <b>OR</b><br/> <math display="block">\text{ext } \angle \text{ of } n\text{-gon} = 180 - 144 = 36</math> <math display="block">n = \frac{360}{36} = 10</math> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|    | b | <div style="display: flex; justify-content: space-between;"> <div> <p>Let <math>k</math> be the sides of the regular polygon</p> <math display="block">(k-2) \times 180 = 81k</math> <math display="block">180k - 360 = 81k</math> <math display="block">99k = 360</math> <math display="block">k = \frac{40}{11}</math> <p>Since <math>k</math> is <b>not a positive integer</b>, there is no regular polygon that can fit at O without gaps.</p> </div> <div> <p><b>OR</b></p> <p>Int angle of regular triangle, 3-gon = <math>60^\circ</math><br/> Int angle of regular quad, 4-gon = <math>90^\circ</math></p> <p>Since <math>60^\circ &lt; 81^\circ &lt; 90^\circ</math>, there is no regular polygon that can fit at O without gaps.</p> </div> </div>                                                                                                                                                 |
|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 18 | a | $\text{Number of blue tokens} = \frac{21-x}{2}$ $P(\text{both blue})$ $= \left( \frac{21-x}{2} \div 21 \right)^2$ $= \left( \frac{21-x}{42} \right)^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|    | b | <div style="display: flex; justify-content: space-between;"> <div> <math display="block">\left( \frac{21-x}{42} \right)^2 = \frac{1}{9}</math> <math display="block">\frac{21-x}{42} = \frac{1}{3} \text{ or } -\frac{1}{3} \text{ (rej.)}</math> <math display="block">21-x = 14</math> <math display="block">x = 7</math> <math display="block">\text{Number of red tokens} = \frac{21-7}{2} + 7 = 14</math> </div> <div> <p><b>OR</b></p> <math display="block">\frac{(21-x)^2}{42^2} = \frac{1}{9}</math> <math display="block">\frac{441 - 42x + x^2}{1764} = \frac{1}{9}</math> <math display="block">x^2 - 42x + 441 = 196</math> <math display="block">x^2 - 42x + 245 = 0</math> <math display="block">(x-7)(x-35) = 0</math> <math display="block">x = 7 \text{ or } x = 35 \text{ (rej.)}</math> <math display="block">\text{Number of red tokens} = \frac{21-7}{2} + 7 = 14</math> </div> </div> |
|    | c | $P(\text{at least one red}) = 1 - P(\text{both blue}) = 1 - \frac{1}{9} = \frac{8}{9}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>19</b> |          | <p><math>P(3,7)</math>, <math>Q(7,9)</math> and <math>R(2,9)</math></p> <p>Notice that <math>Q</math> and <math>R</math> share same <math>y</math>-coordinate, so <math>QR</math> 5 units</p> $PQ = \sqrt{(9-7)^2 + (7-3)^2} = \sqrt{20}$ $PR = \sqrt{(9-7)^2 + (2-3)^2} = \sqrt{5}$ $PQ^2 + PR^2 = 20 + 5$ <p>Since <math>\quad\quad\quad = 25</math></p> $\quad\quad\quad = QR^2$ <p>By the converse of Pythagoras' Theorem, <math>PQR</math> is a right-angled triangle</p> |
| <b>20</b> |          | $x^2 - 8x + k$ $= x^2 - 8x + \left(\frac{-8}{2}\right)^2 - \left(\frac{-8}{2}\right)^2 + k$ $= (x-4)^2 - 16 + k$ <p>Minimum at <math>y = 13</math></p> $k - 16 = 13$ $k = 29$                                                                                                                                                                                                                                                                                                  |
| <b>21</b> | <b>a</b> | <p><math>\triangle AOB</math> and <math>\triangle COD</math>.</p> <p><math>\angle AOB = \angle COD</math> (vert. opp. <math>\angle</math>)</p> <p><math>\angle ABO = \angle CDO</math> (alt. <math>\angle</math>, <math>AB \parallel DC</math>)</p> <p>By the AA Similarity Test, <math>\triangle AOB</math> and <math>\triangle COD</math> are similar.</p>                                                                                                                   |
|           | <b>b</b> | $\frac{12}{20} = \frac{AO}{CO}$ $\frac{3}{5} = \frac{AO}{24 - AO}$ $72 - 3AO = 5AO$ $8AO = 72$ $AO = 9$ $\frac{9}{\sin \angle ABO} = \frac{12}{\sin 82}$ $\sin \angle ABO = \frac{9 \sin 82}{12}$ $\angle ABO = \sin^{-1}(0.7427\dots)$ $= 47.962\dots \approx 48.0$                                                                                                                                                                                                           |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                 |
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| 22 | a | <p>Average Speed</p> $= \frac{\text{Total Distance}}{\text{Total Time}}$ $= \frac{\frac{1}{2}(50 - 10 + 60) \times 6}{60}$ $= 5$                                                                                                                                                                                                                                                |
|    | b |                                                                                                                                                                                                                                                                                               |
|    | c | <p>Distance travelled by Bernard in first 40s</p> $= \frac{1}{2} \times 10 \times 40 = 200$ <p>Distance travelled by Alicia in first 40s</p> $= \frac{1}{2} \times (30 + 40) \times 6 = 210$ <p>Let <math>t</math> be the time Alicia and Bernard meet again. Note: <math>t &gt; 40</math></p> $200 + (t - 40) \times 10 = 210 + (t - 40) \times 6$ $4(t - 40) = 10$ $t = 42.5$ |

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|           |          |                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>23</b> | <b>a</b> | $\tan \angle ACB = \frac{7}{24}$ $\angle ACB = \tan^{-1} \left( \frac{7}{24} \right)$ $= 0.2837941\dots$ $\approx 0.28379$                                                                                                                                                                                                                                                        |
|           | <b>b</b> | $\text{Area of triangle} = \frac{1}{2} \times 7 \times 24 = 84$ $\text{Area of unshaded region } ABE = 84 - \frac{1}{2} \times 24^2 \times 0.28379\dots$ $= 2.26729\dots$ $\approx 2.27$                                                                                                                                                                                          |
|           | <b>c</b> | $AC = \sqrt{7^2 + 24^2} = 25$ $ED = 25 - (25 - 7) - (25 - 24) = 6$ $\angle BAC = \tan^{-1} \left( \frac{24}{7} \right) = 1.28700$ $\text{Arc } EB = 24 \times 0.28379\dots = 6.81106\dots$ $\text{Arc } DB = 7 \times 1.28700\dots = 9.00901\dots$ $\text{Perimeter} = ED + \text{Arc } EB + \text{Arc } DB$ $= 6 + 6.81106\dots + 1.98685\dots$ $= 21.82007\dots$ $\approx 21.8$ |
|           |          |                                                                                                                                                                                                                                                                                                                                                                                   |